DSA Practical No: 07

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Postfix Expression Evalution

Input:

```
#include <stdio.h>
#include <ctype.h>

#define MAX 100

int stack[MAX];
int top = -1;

void push(int value)
{
   stack[++top] = value;
}

int pop()
{
   return stack[top--];
}

int main()
{
   char postfix[MAX];
   int i, a, b, result;
```

```
printf("Enter postfix expression: ");
scanf("%99s", postfix);

for (i = 0; postfix[i] != '\0'; i++)
{
   if (isdigit((unsigned char)postfix[i]))
   {
      push(postfix[i] - '0');
   }
   else
   {
      b = pop();
      a = pop();

      switch (postfix[i])
      {
         case '+':
            result = a + b;
            break;

         case '-':
            result = a - b;
            break;

         case '*':
            result = a * b;
            break;
```

```
            case '/':
               if (b == 0)
               {
                  printf("Division by zero error\n");
                  return 1;
               }
               result = a / b;
               break;

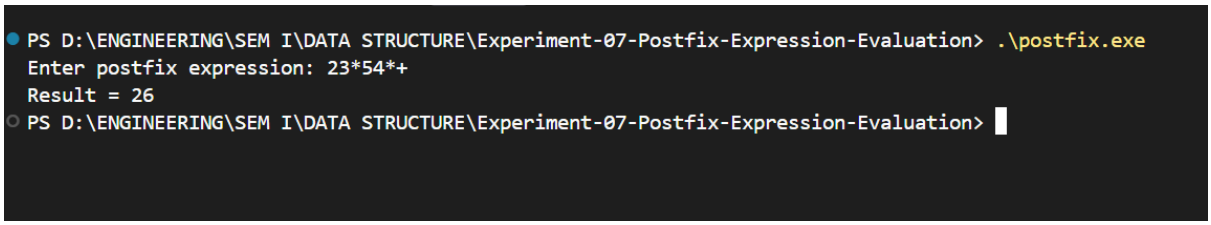
            default:
               printf("Invalid operator\n");
               return 1;
         }

         push(result);
      }
   }

   if (top == 0)
      printf("Result = %d\n", pop());
   else
      printf("Invalid postfix expression\n");

   return 0;
}
```

Output:


PS D:\ENGINEERING\SEM I\DATA STRUCTURE\Experiment-07-Postfix-Expression-Evaluation> .\postfix.exe
Enter postfix expression: 23*54*+
Result = 26
PS D:\ENGINEERING\SEM I\DATA STRUCTURE\Experiment-07-Postfix-Expression-Evaluation>